RESULTS & CONCLUSIONS
Defense-in-Depth Works

IoT Security Demonstration: Vulnerable vs. Secure System Comparison

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ATTACK OUTCOME COMPARISON

VULNERABLE SYSTEM
Pi #7 — 192.168.50.100

SECURITY CONTROLS
- ✗ NO authentication
- ✗ NO input validation
- ✗ NO sensor cross-check
- ✗ NO anomaly detection
- ✗ NO SIEM logging

ATTACK RESULTS
- 100% attack success rate
- 150+ requests accepted
- 0 detections triggered
- < 20 seconds to failure

PHYSICAL OUTCOME
- 28.7 PSI achieved (dangerous)
- +3.7 PSI overpressure
- Safe limit exceeded: 25 PSI
- Pump ran uncontrolled

DETECTION & RESPONSE
- Suricata: traffic logged only
- Forensic value only
- Too late to prevent damage
- No real-time protection

SECURE SYSTEM
Pi #6 — 10.30.0.25

SECURITY CONTROLS
- ✓ Rate limiting (2 Hz max)
- ✓ Input validation (0–100 PSI)
- ✓ Sensor cross-check (±1 PSI)
- ✓ Anomaly detection (5 PSI/sec)
- ✓ Wazuh SIEM logging

ATTACK RESULTS
- 0% attack success rate
- 150+ requests rejected
- 150+ detections triggered
- < 1 second detection latency

PHYSICAL OUTCOME
- 25.0 PSI maintained (safe ✓)
- 0 PSI overpressure
- At target: 25 PSI
- Pump remained off

DETECTION & RESPONSE
- Wazuh: all attacks logged
- Real-time detection (< 1s)
- Forensic data preserved
- Active defense successful ✓

COMPARATIVE METRICS

100% vs 0%
Attack Success
Vulnerable: 100%
Secure: 0%

0 vs 150+
Detections
Vulnerable: 0
Secure: 150+

28.7 vs 25.0
Final PSI
Vulnerable: 28.7
Secure: 25.0

< 20s vs ∞
Time to Failure
Vulnerable: < 20s
Secure: Never

0 vs 5
Security Layers
Vulnerable: 0
Secure: 5

7 vs 0
Critical Flaws
Vulnerable: 7
Secure: 0

KEY LESSONS & RECOMMENDATIONS

LESSONS LEARNED

1. DEFENSE IN DEPTH WORKS
- Multiple overlapping controls provide resilience
- Single-layer security fails under attack

2. SENSOR VALIDATION CRITICAL
- Cross-check network data vs physical sensors
- Blind trust in network data = total compromise

3. REAL-TIME DETECTION MATTERS
- < 1s detection prevents physical damage
- Post-hoc forensics cannot prevent incidents

4. LOGGING ENABLES FORENSICS
- SIEM logging provides attack attribution
- Essential for incident response
- Threat intelligence and timeline analysis

RECOMMENDATIONS

FOR MEDICAL DEVICE MAKERS
- ✓ Implement sensor redundancy
- ✓ Validate all network inputs
- ✓ Deploy anomaly detection + SIEM

FOR HOSPITAL IT / SECURITY
- ✓ Segment IoT networks (VLANs)
- ✓ Deploy IDS + monitor all devices
- ✓ Regular security assessments

FOR REGULATORY BODIES
- ✓ Mandate sensor validation
- ✓ Require security logging
- ✓ Enforce defense-in-depth
- ✓ Follow NIST / IEC 62443 / Zero Trust

PROJECT ACHIEVEMENTS

INFRASTRUCTURE BUILT
- 9 virtual machines
- 7 Raspberry Pi devices
- VLAN segmentation
- 14-agent SIEM (Wazuh)
- IDS/SIEM pipeline
- SSO (Keycloak)
- Secrets mgmt (Vault)
- 53M+ data points

SECURITY DEMONSTRATED
- Sensor spoofing attack
- Multi-layer defense
- Real-time detection
- SIEM correlation
- Physical consequence demo
- 100% vs 0% success rates
- Attack vs defense comparison
- Forensic logging

STANDARDS APPLIED
- HL7 FHIR R4 (health data)
- HIPAA controls
- TLS/SSL encryption
- OAuth2 (Keycloak)
- Network segmentation
- Defense-in-depth
- Least privilege
- NIST / IEC 62443

ACKNOWLEDGMENTS

University of Cincinnati | College of Engineering and Applied Science | Electrical & Computer Engineering

Advisor: Rob Kuschel, GE Aerospace CTH | Special thanks: UC EECE Faculty, GE Aerospace, open-source security community

Technologies: Raspberry Pi • Python • PostgreSQL • Grafana • Node-RED • Suricata IDS • Wazuh SIEM • Keycloak • HashiCorp Vault • MQTT • HL7 FHIR

Standards: NIST Cybersecurity Framework • IEC 62443 • Zero Trust • HIPAA • OAuth2 • TLS/SSL